multiclass logistic regression

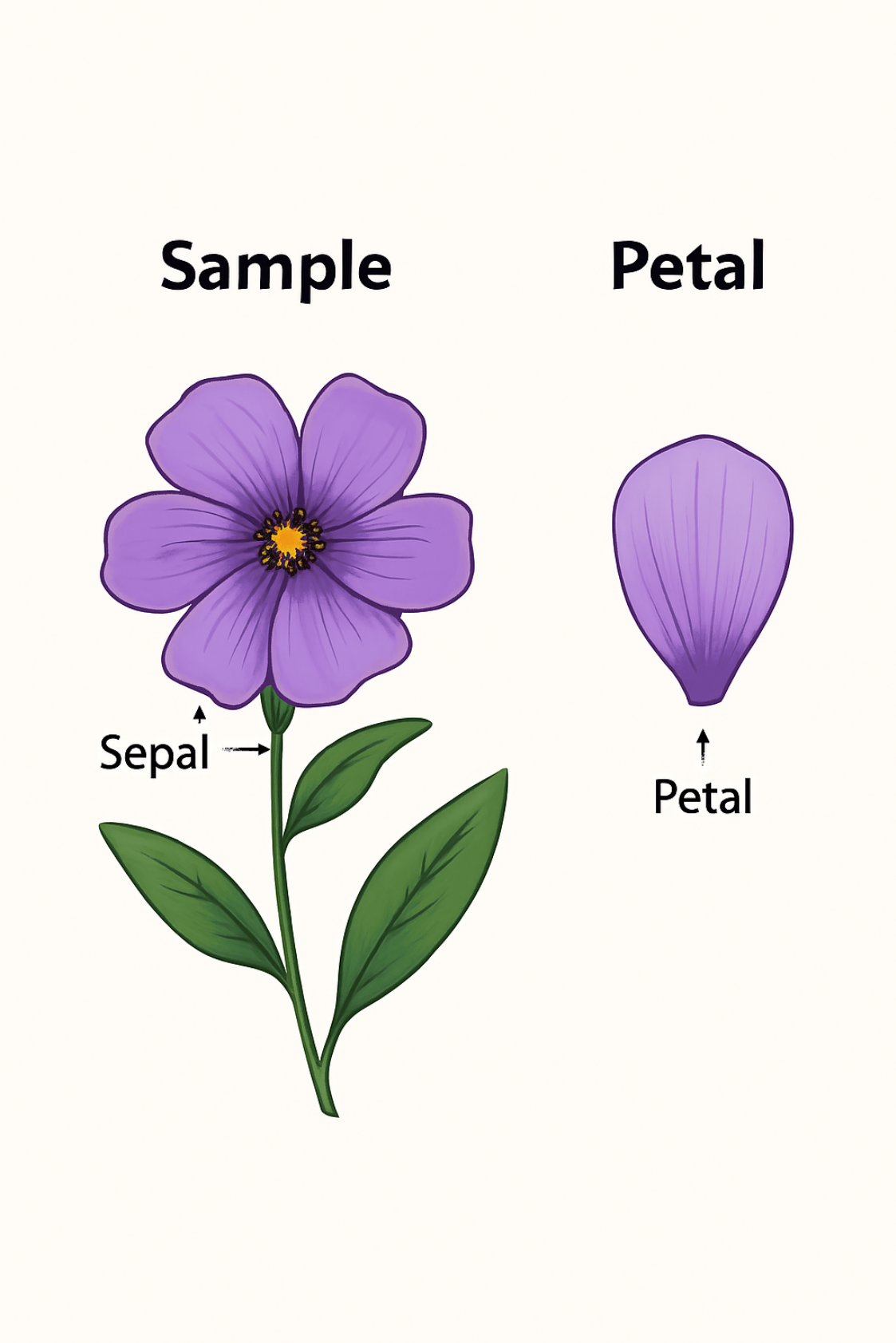
April 2, 2026

[2]:
```
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns
```

[3]:
```
from sklearn.datasets import load_iris
```

[4]:
```
iris=load_iris(as_frame=True)
df=iris.frame
df.head()
```

[4]:
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target
0       0
1       0
2       0
3       0
4       0
```


Sample
Petal
Sepal
Petal

[5]:
```python
df.target.unique()
```

[5]:
```
array([0, 1, 2])
```

[6]:
```python
iris.target_names
```

[6]:
```
array(['setosa', 'versicolor', 'virginica'], dtype='<U10')
```

[7]:
```python
df.shape
```

[7]:
```
(150, 5)
```

[8]:
```python
print(iris.DESCR)
```

```
.. _iris_dataset:

Iris plants dataset
--------------------

**Data Set Characteristics:**

:Number of Instances: 150 (50 in each of three classes)
:Number of Attributes: 4 numeric, predictive attributes and the class
:Attribute Information:
    - sepal length in cm
    - sepal width in cm
    - petal length in cm
    - petal width in cm
    - class:
            - Iris-Setosa
            - Iris-Versicolour
            - Iris-Virginica

:Summary Statistics:

============== ==== ==== ======= ===== ====================
                Min  Max   Mean    SD   Class Correlation
============== ==== ==== ======= ===== ====================
sepal length:   4.3  7.9   5.84   0.83    0.7826
sepal width:    2.0  4.4   3.05   0.43   -0.4194
petal length:   1.0  6.9   3.76   1.76    0.9490  (high!)
petal width:    0.1  2.5   1.20   0.76    0.9565  (high!)
============== ==== ==== ======= ===== ====================

:Missing Attribute Values: None
:Class Distribution: 33.3% for each of 3 classes.
:Creator: R.A. Fisher
:Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)
```

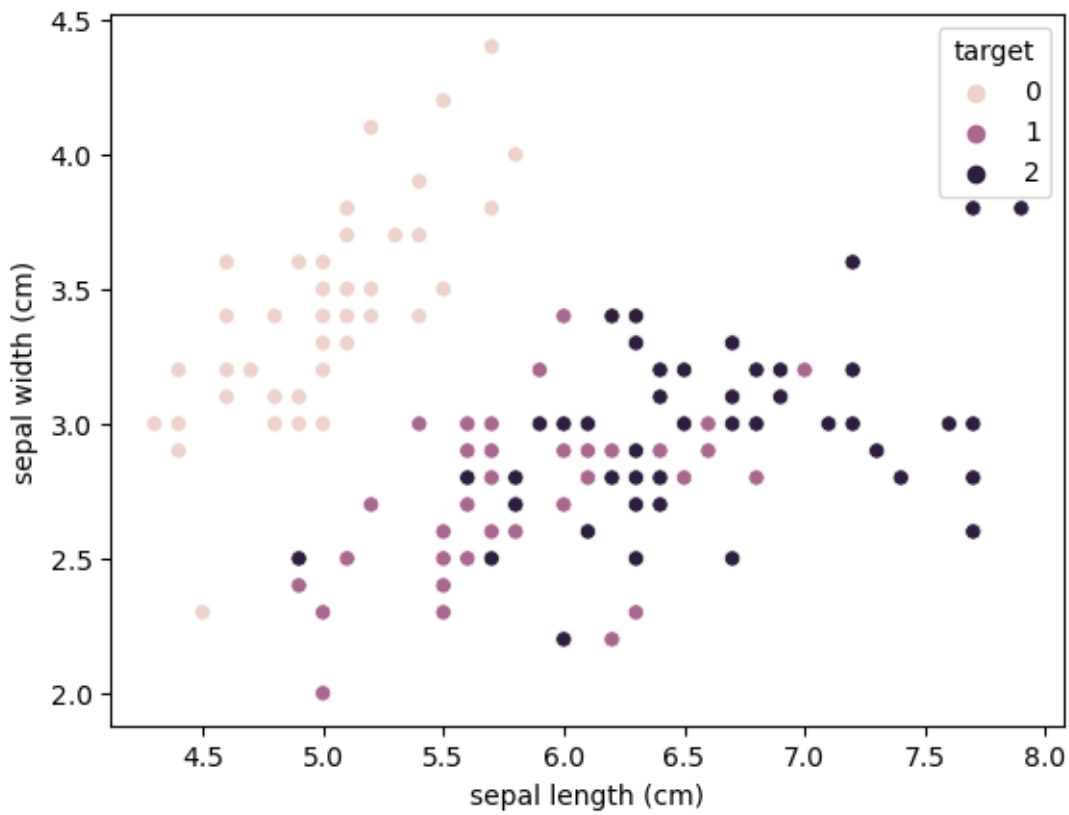
```
:Date: July, 1988

The famous Iris database, first used by Sir R.A. Fisher. The dataset is taken
from Fisher's paper. Note that it's the same as in R, but not as in the UCI
Machine Learning Repository, which has two wrong data points.

This is perhaps the best known database to be found in the
pattern recognition literature.  Fisher's paper is a classic in the field and
is referenced frequently to this day.  (See Duda & Hart, for example.)  The
data set contains 3 classes of 50 instances each, where each class refers to a
type of iris plant.  One class is linearly separable from the other 2; the
latter are NOT linearly separable from each other.

.. dropdown:: References

  - Fisher, R.A. "The use of multiple measurements in taxonomic problems"
    Annual Eugenics, 7, Part II, 179-188 (1936); also in "Contributions to
    Mathematical Statistics" (John Wiley, NY, 1950).
  - Duda, R.O., & Hart, P.E. (1973) Pattern Classification and Scene Analysis.
    (Q327.D83) John Wiley & Sons.  ISBN 0-471-22361-1.  See page 218.
  - Dasarathy, B.V. (1980) "Nosing Around the Neighborhood: A New System
    Structure and Classification Rule for Recognition in Partially Exposed
    Environments".  IEEE Transactions on Pattern Analysis and Machine
    Intelligence, Vol. PAMI-2, No. 1, 67-71.
  - Gates, G.W. (1972) "The Reduced Nearest Neighbor Rule".  IEEE Transactions
    on Information Theory, May 1972, 431-433.
  - See also: 1988 MLC Proceedings, 54-64.  Cheeseman et al"s AUTOCLASS II
    conceptual clustering system finds 3 classes in the data.
  - Many, many more …
```

[9]:
```
#  Data Visualization
```

[10]:
```
import seaborn as sns
```

[11]:
```
sns.scatterplot(data=df, x="sepal length (cm)", y="sepal width (cm)",␣
↪hue="target")
```

[11]:
```
<Axes: xlabel='sepal length (cm)', ylabel='sepal width (cm)'>
```



[12]:
```
df["target_name"]= df["target"].map({0:"setosa",1:"versicolor",2:"virginica"})
```

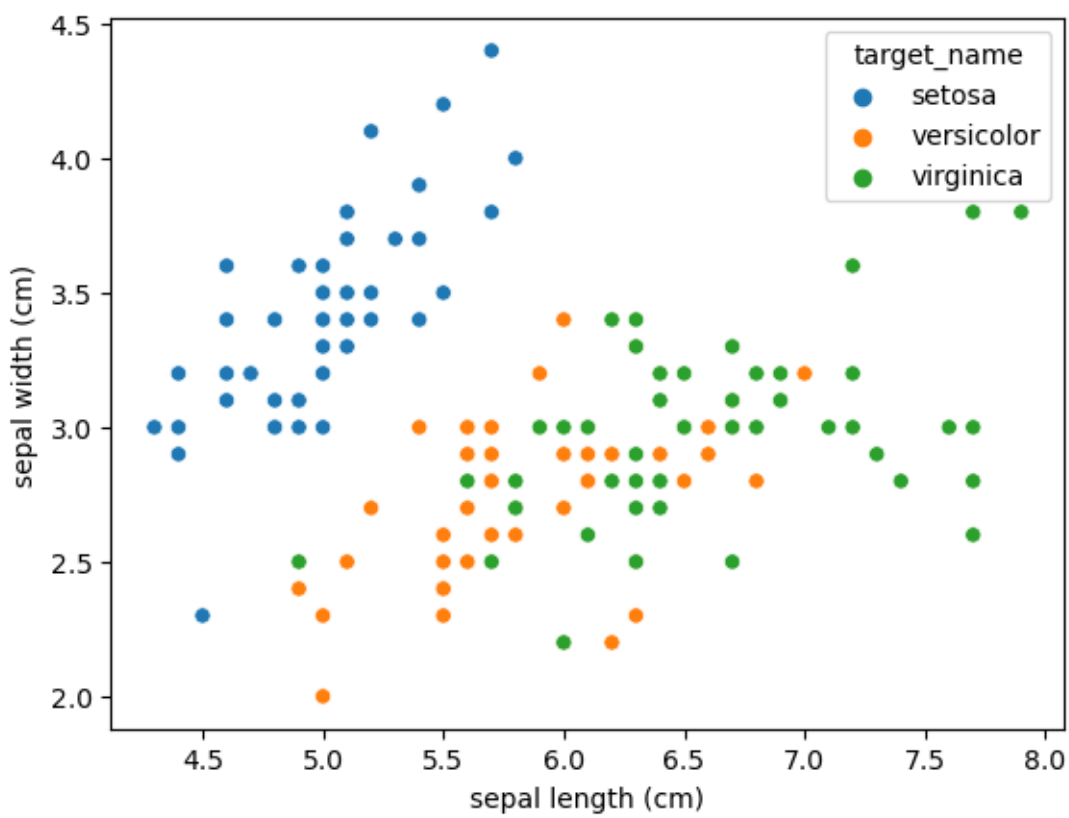
[13]:
```
df
```

[13]:
```
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                  5.1               3.5                1.4               0.2
1                  4.9               3.0                1.4               0.2
2                  4.7               3.2                1.3               0.2
3                  4.6               3.1                1.5               0.2
4                  5.0               3.6                1.4               0.2
..                 …                 …                  …                 …
145                6.7               3.0                5.2               2.3
146                6.3               2.5                5.0               1.9
147                6.5               3.0                5.2               2.0
148                6.2               3.4                5.4               2.3
149                5.9               3.0                5.1               1.8

     target target_name
0         0      setosa
1         0      setosa
2         0      setosa
```

```
3          0      setosa
4          0      setosa
..       …          …
145        2   virginica
146        2   virginica
147        2   virginica
148        2   virginica
149        2   virginica

[150 rows x 6 columns]
```

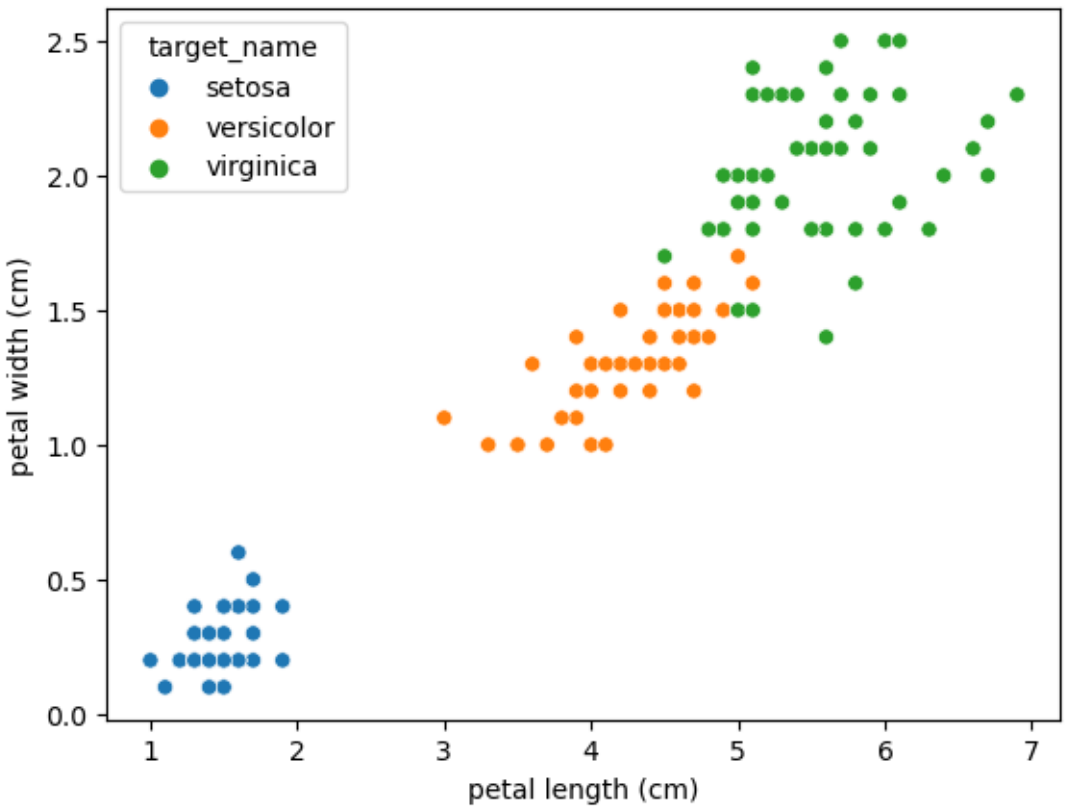
[14]:
```
sns.scatterplot(data=df, x="sepal length (cm)", y="sepal width (cm)",␣
↪hue="target_name")
```

[14]: <Axes: xlabel='sepal length (cm)', ylabel='sepal width (cm)'>



[15]:
```
sns.scatterplot(data=df, x="petal length (cm)", y="petal width (cm)",␣
↪hue="target_name")
```

[15]: <Axes: xlabel='petal length (cm)', ylabel='petal width (cm)'>



[16]:
```
X=df[["sepal length (cm)","sepal width (cm)","petal length (cm)","petal width␣
↪(cm)"]]
y=df["target"]
```

[17]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
```

[18]:
```
lm=LogisticRegression(max_iter=200)
```

[19]:
```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.
↪3,random_state=42)
```

[20]:
```
lm.fit(X_train,y_train)
```

[20]: LogisticRegression(max_iter=200)

[21]:
```
lm.score(X_test,y_test)
```

[21]: 1.0

[22]:
```
y_pred=lm.predict(X_test)
```

[23]:
```
from sklearn.metrics import accuracy_score
accuracy_score(y_test,y_pred)
```

[23]: 1.0

[24]:
```
from sklearn.metrics import confusion_matrix,classification_report
```

[25]:
```
cm=confusion_matrix(y_test,y_pred)
```

[26]:
```
cm
```

[26]:
```
array([[19,  0,  0],
       [ 0, 13,  0],
       [ 0,  0, 13]], dtype=int64)
```

[27]:
```
cr=classification_report(y_test,y_pred)
```

[28]:
```
print(cr)
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      1.00      1.00        13
           2       1.00      1.00      1.00        13

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45
```

[33]:
```
lm.predict([[6,4,2,1]])
```

```
C:\Users\saura\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739:
UserWarning: X does not have valid feature names, but LogisticRegression was
fitted with feature names
  warnings.warn(
```

[33]:
```
array([0])
```

[32]:
```
df.head(5)
```

[32]:
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target target_name
```

```
0       0      setosa
1       0      setosa
2       0      setosa
3       0      setosa
4       0      setosa
```

[ ]:

```

```